

CoRoT-1b

R-2618

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_150203 · created 2026-08-28T23:12:59+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457056.6678 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.076 +/- 0.041
Transit depth (Rp/Rs)^2	0.57 +/- 0.62 [%]
Orbital Inclination (inc)	89.79 +/- 2.8
Airmass coefficient 1 (a1)	1.31 +/- 0.61
Airmass coefficient 2 (a2)	-0.15 +/- 0.36
Scatter in the residuals of the lightcurve fit is	48.55 %
Best Comparison Star	#2 - [492, 412]
Optimal Aperture	5.43
Optimal Annulus	14.36
Transit Duration (day)	0.1058 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.076 ± 0.041 vs 0.1388 (deviation 1.5σ)
Duration fit vs published	0.1058 d vs 0.1046 d (deviation 0.22σ)
Mid-time O–C	-7.4 min
Depth SNR	1.8
Residual scatter	48.55 %

Light curve

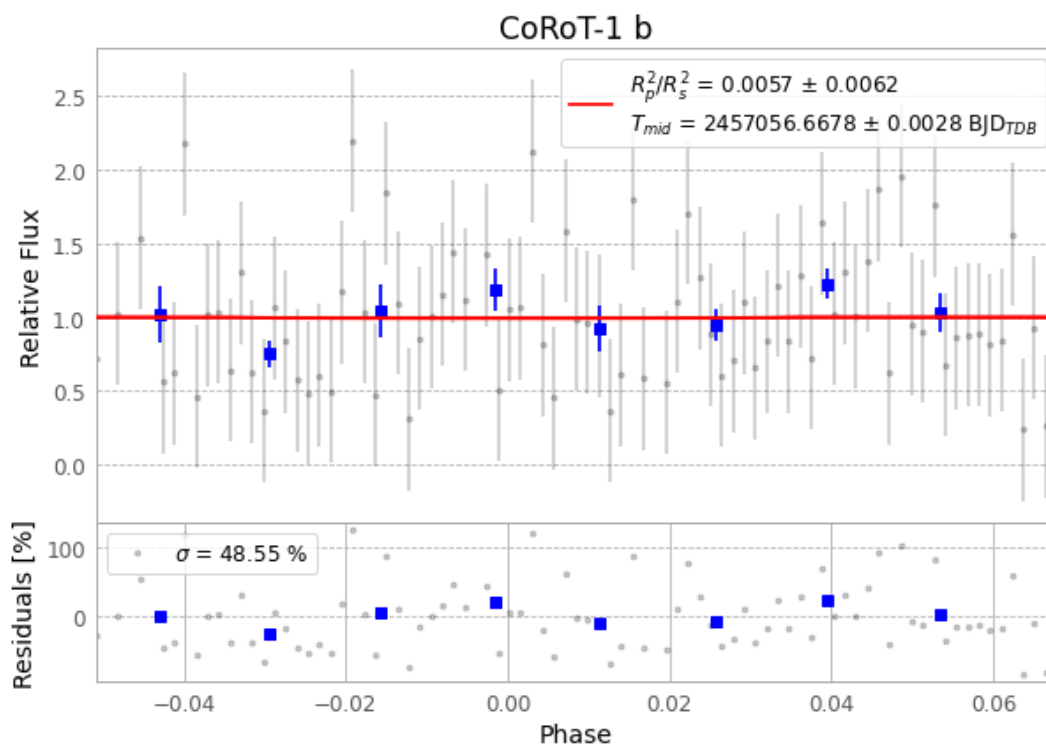


Figure 1 — FinalLightCurve_CoRoT-1 b_2015-02-02.png (SHA-256 ba4ccb6d3e19b27e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [420, 355], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 189bc7e2fa335eda...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

87 FITS frames (57 MB), CoRoT-1150203020312.FITS → CoRoT-1150203062112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-150203.log (831883e7d745...), FinalLightCurve_CoRoT-1 b_2015-02-02.png (ba4ccb6d3e19...), FinalLightCurve_CoRoT-1 b_2015-02-02.csv (94e89f14f7b4...)

Compute resources

Wall time 113.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 23:13Z.